

PROJECT DESIGN DOCUMENT (PDD)

Amazon Sales Dashboard Analysis

Technology: Microsoft Power BI

Data Source: Amazon-style Sales Dataset

Dataset Size: 6,000 rows × 10 columns

Dashboard Pages: 3

1. Project Overview

The **Amazon Sales Dashboard Analysis** is an interactive Business Intelligence project developed using Microsoft Power BI.

The project transforms raw Amazon-style sales transaction data into a clean, interactive dashboard that helps users understand sales performance, product performance, regional trends, customer activity, and payment-method usage.

The dashboard is designed to allow business users to move from raw transactional data to meaningful business insights through data cleaning, calculated columns, DAX measures, KPIs, charts, filters, and interactive navigation.

2. Problem Statement

Raw sales data may contain duplicate records, missing values, inconsistent text formats, invalid quantities, and incorrect or incomplete sales values.

Analyzing such data directly can result in inaccurate sales figures and misleading business insights.

Therefore, the project aims to:

- Clean and validate the raw sales dataset.
- Remove duplicate and invalid records.
- Handle missing values appropriately.
- Standardize inconsistent text data.
- Recalculate sales values accurately.
- Build reusable DAX measures.
- Develop an interactive three-page Power BI dashboard.

- Identify important sales, product, regional, customer, and payment trends.

3. Project Objectives

The main objectives of the project are:

1. Analyze overall sales performance.
2. Identify high-performing product categories.
3. Identify top-performing products.
4. Analyze monthly sales trends.
5. Compare sales across regions.
6. Analyze payment-method performance.
7. Understand order and customer distribution.
8. Compare revenue with units sold.
9. Create interactive KPIs and visualizations.
10. Provide data-driven insights for business decision-making.

4. Key Business Questions

The dashboard answers the following questions:

- Which categories generate the highest revenue?
- Which products contribute the most to sales?
- How does sales performance change month over month?
- Which regions generate the highest sales and orders?
- Which payment methods are used most frequently?
- How are sales distributed across regions?
- Which categories have high sales but comparatively lower unit volume?
- How many unique customers are represented in the dataset?
- How does data cleaning affect the reliability of the analysis?

5. Dataset Overview

The project uses an Amazon-style sales dataset containing **6,000 raw records and 10 columns** covering January 2023 to December 2024.

Dataset Columns

Column	Data Type	Description
Order_ID	Text	Unique order identifier
Order_Date	Date	Date on which the order was placed
Category	Text	Product category
Product_Name	Text	Name of the product
Quantity	Whole Number	Number of units purchased
Price	Decimal	Unit price
Sales	Decimal	Sales/revenue value
Region	Text	Customer/order region
Customer_Name	Text	Customer name
Payment_Method	Text	Payment method used

6. Data Quality Challenges

The raw dataset contained:

- 6,000 total rows
- 180 duplicate rows
- 508 rows containing missing values
- Inconsistent region formatting
- Invalid/non-positive quantity values
- Potentially inconsistent sales values

After cleaning, the dataset contained **5,820 valid unique orders**.

7. Data Cleaning Approach

Power Query was used to prepare the dataset before analysis.

The major cleaning steps were:

1. Promote the first row as headers.
2. Assign appropriate data types.
3. Trim unnecessary spaces.
4. Standardize region names.
5. Remove duplicate records.
6. Handle missing values.
7. Remove invalid quantity records.
8. Recalculate sales using $\text{Quantity} \times \text{Price}$.
9. Load the cleaned dataset into the Power BI data model.

These steps help ensure that the dashboard calculations are based on consistent and reliable data.

8. Dashboard Structure

The dashboard consists of three interactive pages.

Page 1 — Sales Overview

Focuses on overall sales performance.

Key KPIs:

- Total Sales
- Total Orders
- Average Order Value
- Units Sold

Visuals:

- Monthly Sales Trend
- Sales by Payment Method
- Sales by Region

Page 2 — Product & Category Analysis

Focuses on product and category performance.

Key KPIs:

- Number of Categories
- Number of Products
- Top Category
- Top Product

Visuals:

- Sales by Category
- Top 5 Products by Sales
- Units Sold by Category

Page 3 — Regional & Customer Analysis

Focuses on geographic and customer-level activity.

Key KPIs:

- Unique Customers
- Number of Regions
- Top Region
- Sales per Customer

Visuals:

- Orders by Region
- Regional Sales Share
- Orders by Payment Method

9. Key KPIs

The dashboard provides the following major performance indicators:

KPI	Purpose
Total Sales	Measures overall revenue
Total Orders	Measures number of unique orders
Average Order Value	Measures average revenue per order
Total Quantity	Measures units sold
Total Customers	Measures unique customers
Sales MoM % Change	Measures monthly sales growth/change
Top Category Sales	Identifies the highest-revenue category
% of Total Sales	Measures contribution to total revenue

10. Expected Business Value

The dashboard provides a single interactive reporting environment for:

- Sales monitoring
- Product performance analysis
- Category analysis
- Regional analysis
- Customer analysis
- Payment-method analysis
- Trend identification
- Data-quality monitoring

It reduces the need for manual Excel-based reporting and provides a more consistent way to analyze sales data.

11. Project Deliverables

The final project includes:

- Raw sales dataset
- Cleaned Power Query dataset

- Power BI data model
- Calculated columns
- DAX measures
- Three-page interactive dashboard
- KPI cards
- Charts and visualizations
- Business insights and recommendations
- Project documentation

12. Conclusion

The Amazon Sales Dashboard Analysis demonstrates a complete Business Intelligence workflow:

Raw Data → Data Cleaning → Data Transformation → Data Modeling → DAX → Visualization → Business Insights

The project converts raw transactional data into an interactive dashboard that provides a clear view of sales, products, categories, regions, customers, and payment methods.

The final solution demonstrates practical skills in **Power BI, Power Query, DAX, data cleaning, data visualization, KPI development, and business analysis.**